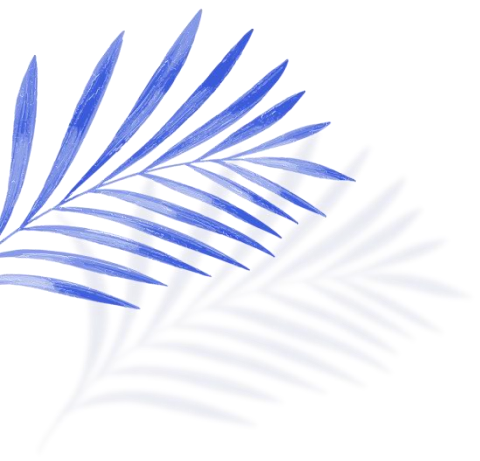




2025.5

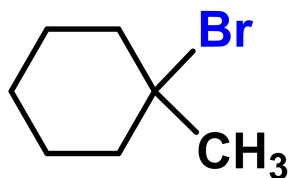
期末考试复习



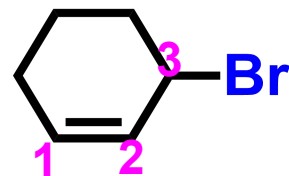
一、命名题 ($4 \times 2 = 8$ 分) ?

- 降低整体难度
- 不考立体化学

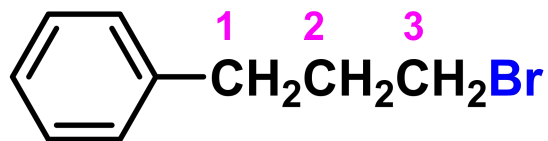
卤代烃



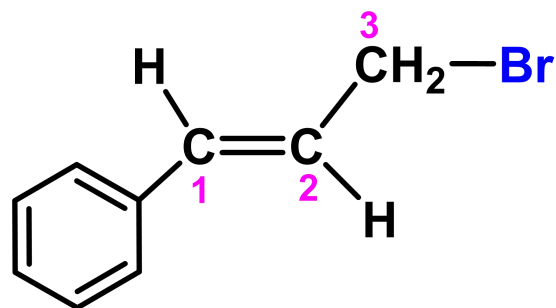
1-溴-1-甲基环己烷



3-溴环己烯

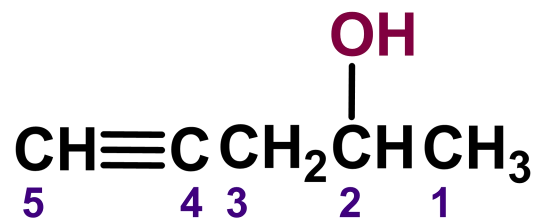


(3-溴丙基)苯

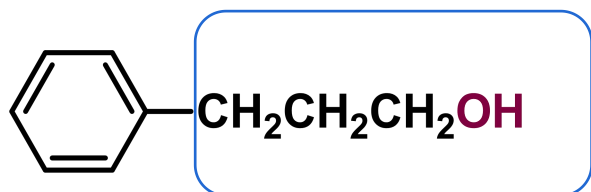


(*E*)-(3-溴丙-1-烯-1-基)苯

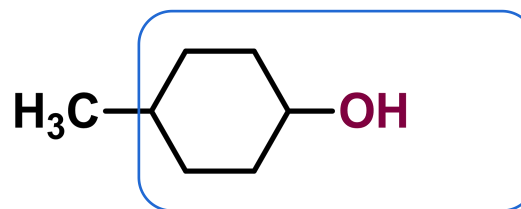
醇



戊-4-炔-2-醇

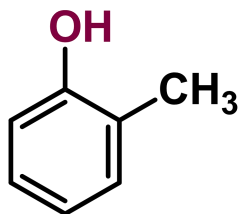


3-苯基丙-1-醇

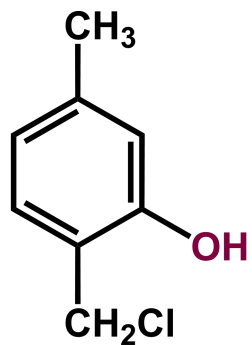


4-甲基环己醇

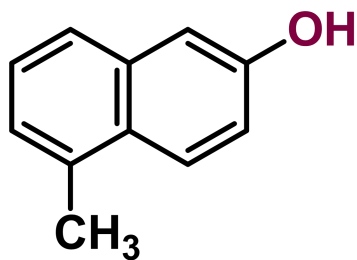
酚



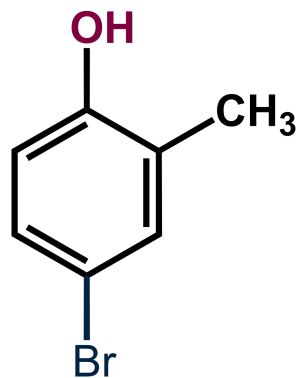
2-甲基苯酚 (邻甲苯酚)



2-(chloromethyl)-5-ethylphenol

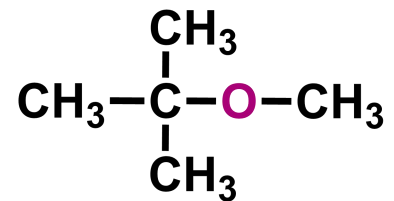


5-甲基萘-2-酚

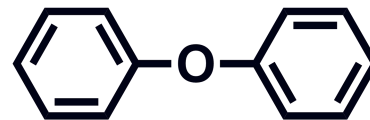


4-溴-2甲基苯酚

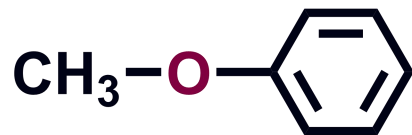
醚



叔丁基甲基醚



二苯(基)醚



苯甲醚

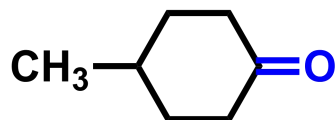


四氢呋喃 (氧杂环戊烷)

醛酮

羰基在环内--环某酮；

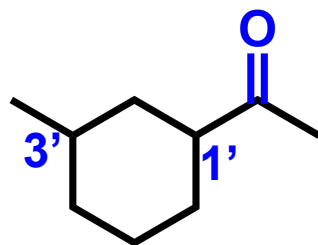
羰基在环外--环为取代基，脂环族醛--烃名称加醛



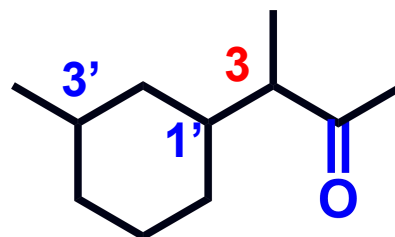
4-甲基环己酮



4-甲基环己基甲醛

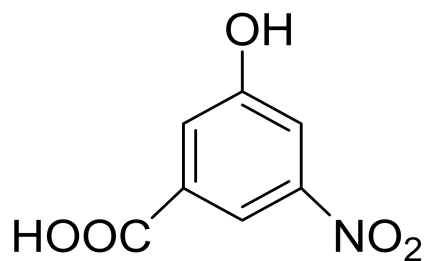


1-(3-甲基环己基)乙酮

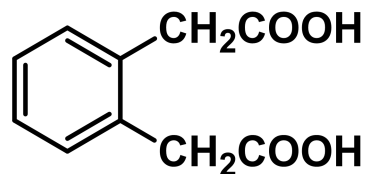


3-(3-甲基环己基)丁-2-酮

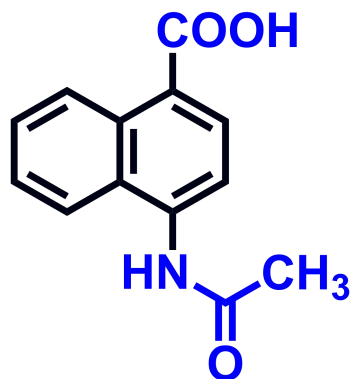
羧酸



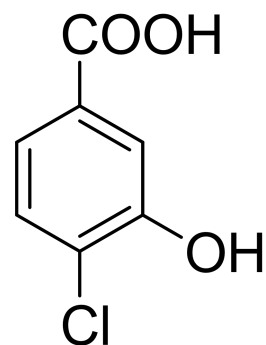
3-羟基-5-硝基-苯甲酸



苯-1,2-二乙酸

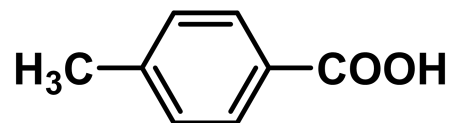


4-乙酰氨基萘-1-甲酸

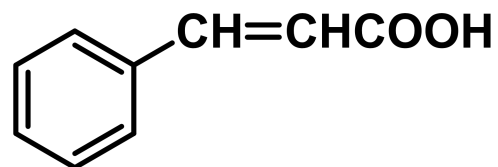


4-氯-3-羟基-苯甲酸

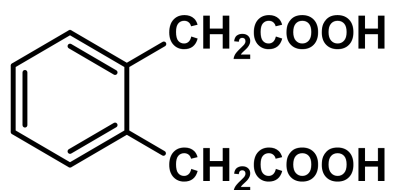
酸



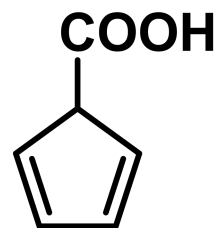
4-甲基苯甲酸



3-苯基丙烯酸

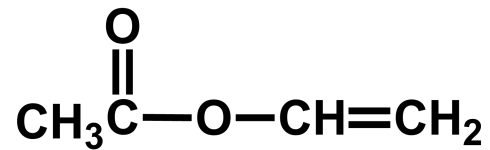


苯-1,2-二乙酸

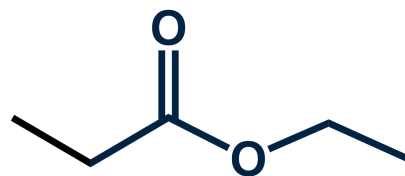


环戊-2,4-二烯甲酸

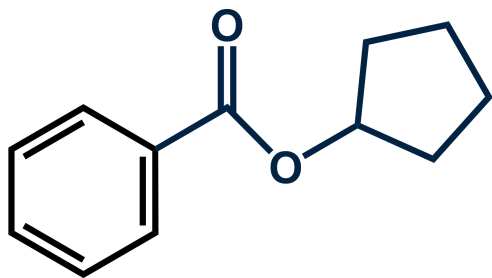
酯



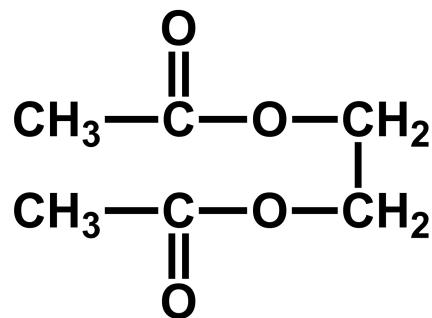
乙酸乙烯酯



丙酸乙酯

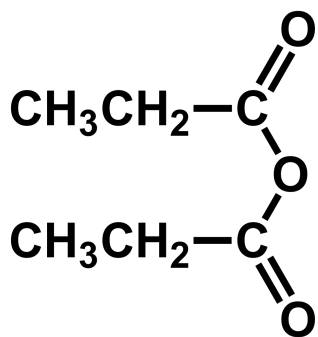


环戊基苯甲酸酯

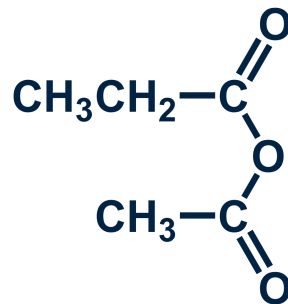


乙二醇二乙酸酯

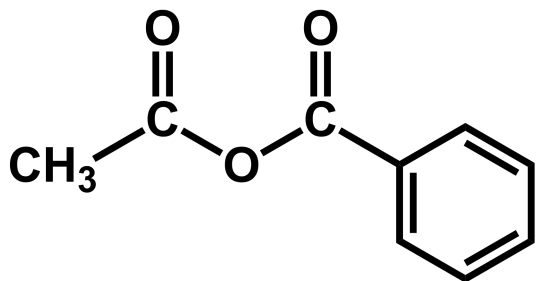
酸酐



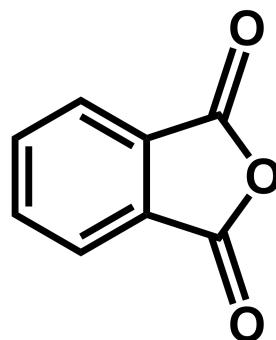
丙酸酐



乙酸甲酸酐

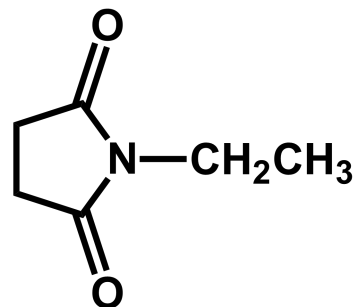


乙酸苯甲酸酐

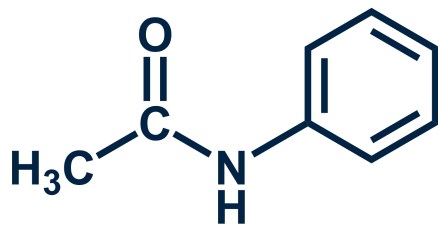


苯-1,2-二甲酸酐

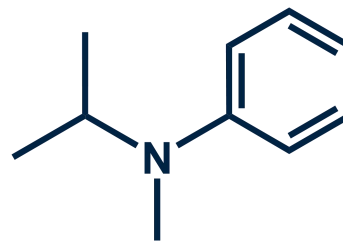
酰胺



N-乙基丁二酰亚胺

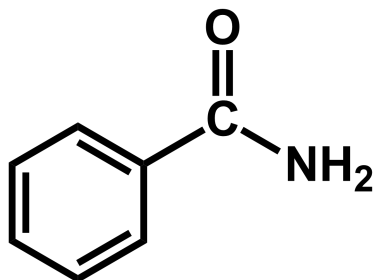


N-苯基乙酰胺

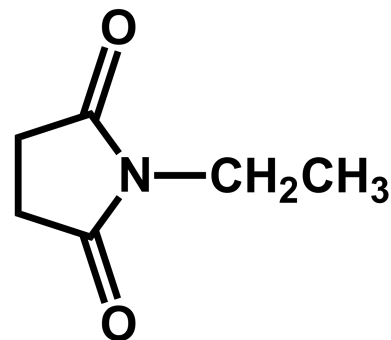


N-异丙基-N甲基-本胺

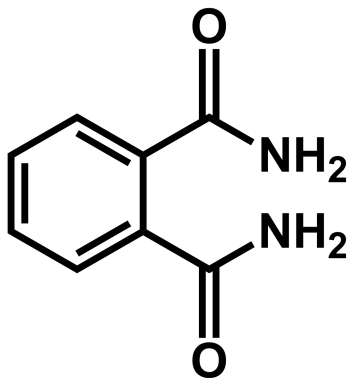
酰胺



苯甲酰胺

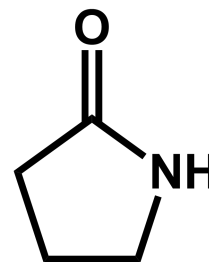


N-乙基丁二酰亚胺



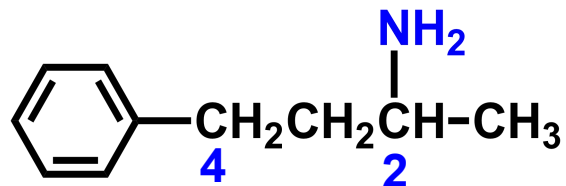
苯-1,2-二甲酰胺

邻苯二甲酰胺

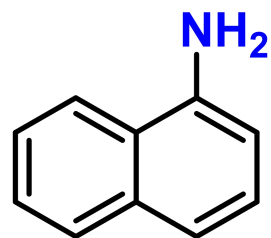


丁-4-内酰胺

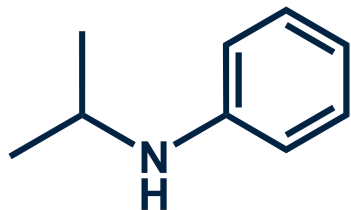
胺



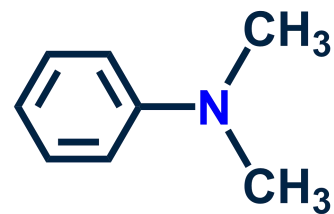
4-苯基丁-2-胺



萘-1-胺

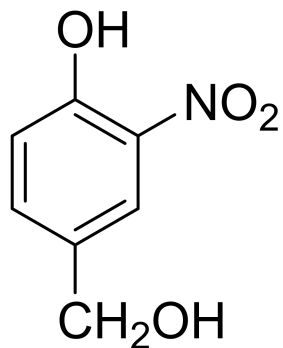
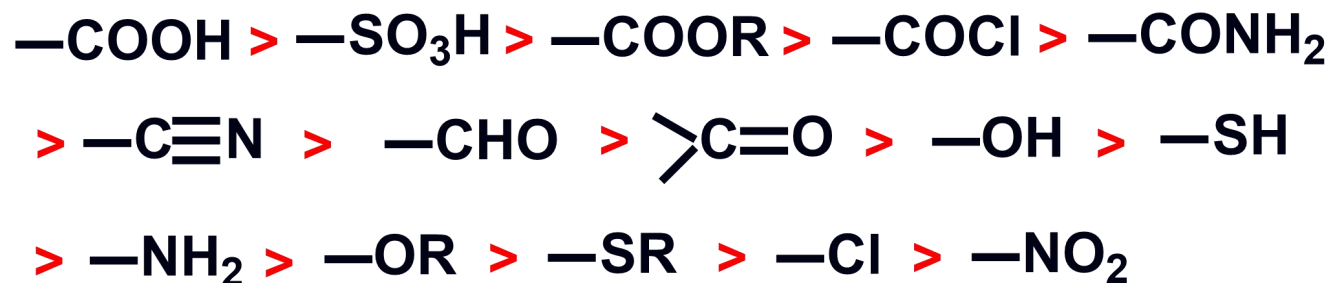


N-异丙基苯胺

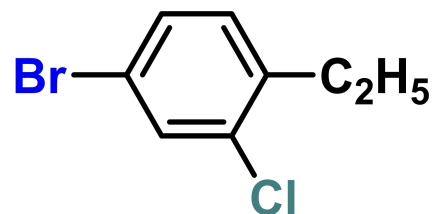


N,N-二甲基苯胺

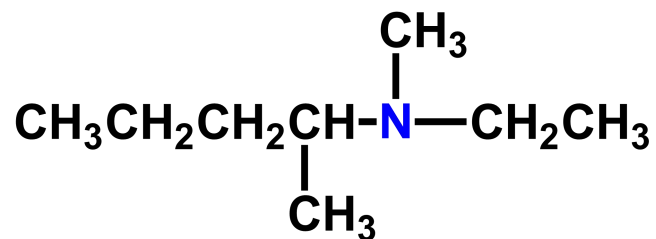
多取代芳烃



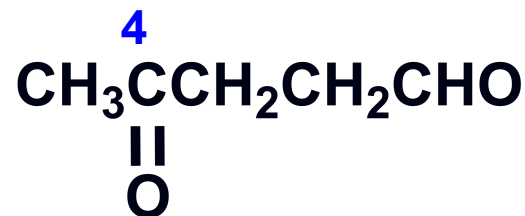
4-羧甲基-2-硝基-苯酚



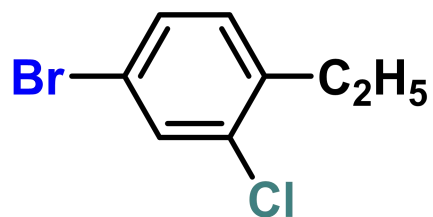
4-溴-2-氯-1-乙基苯



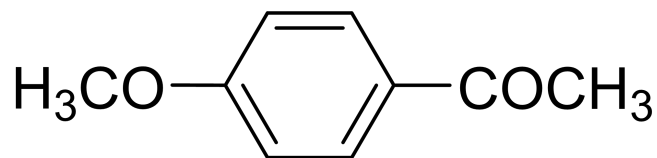
***N*-乙基-*N*-甲基戊-2-胺**



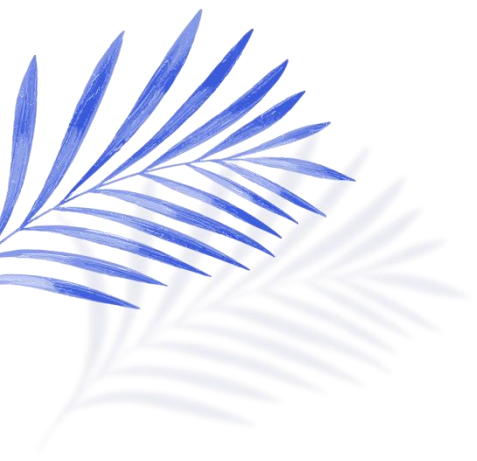
4-氧亚基戊醛



4-溴-2-氯-1-乙基苯



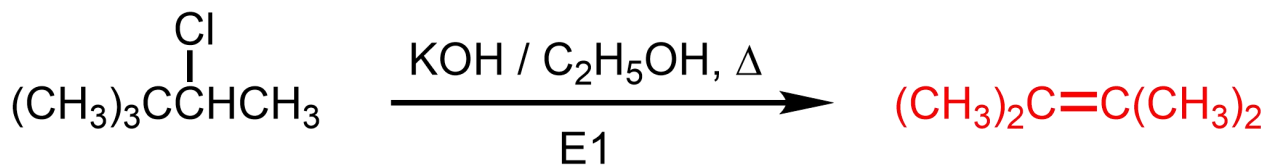
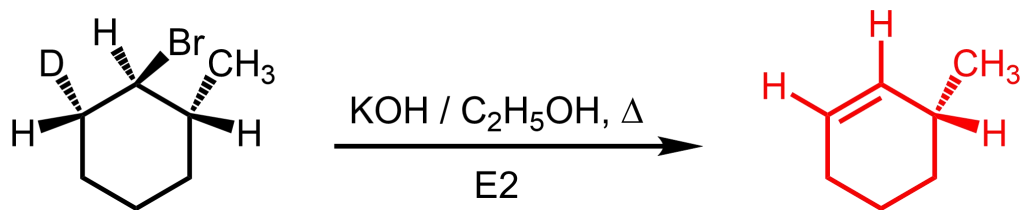
4-甲氧基-苯乙酮



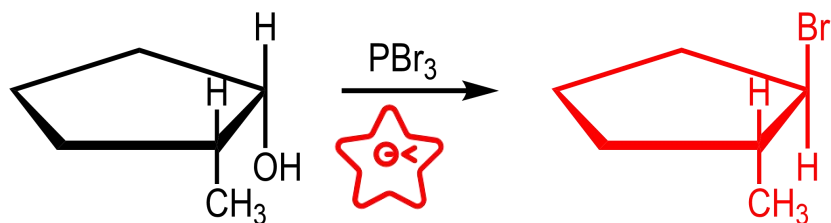
二、完成反应方程式 ($12 \times 2 = 24$ 分)

- ① 降低难度，基本是一步反应
- ② 注意反应条件

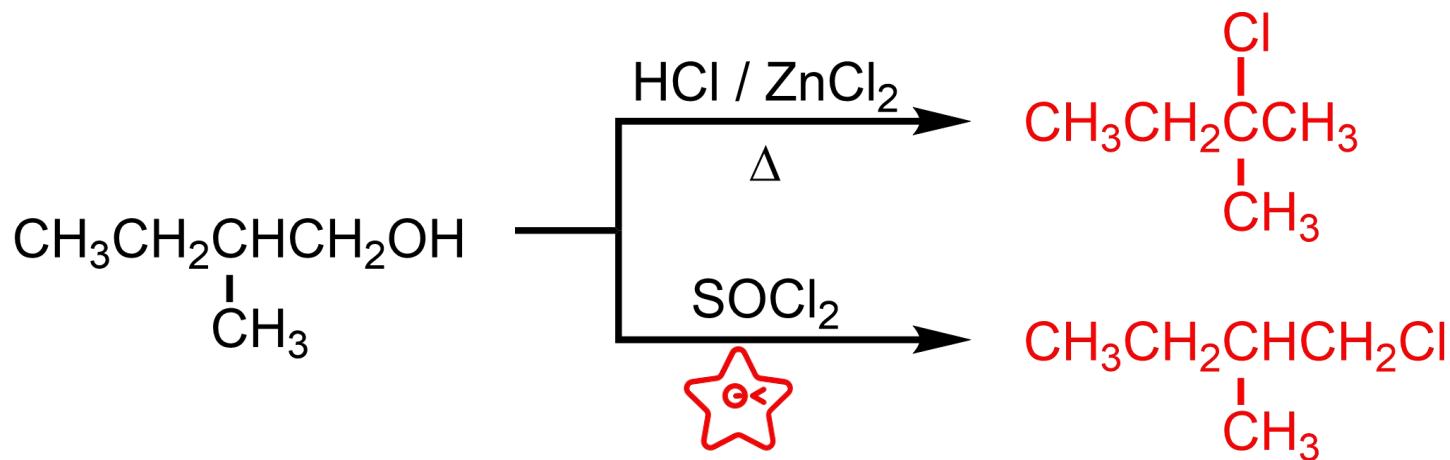
卤代烃:



醇的卤代



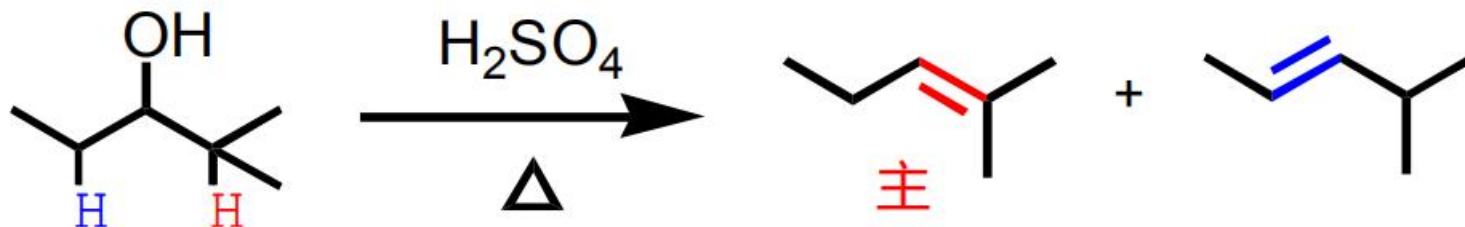
该反应按 $\text{S}_{\text{N}}2$ 进行，不发生重排，构型发生转化。



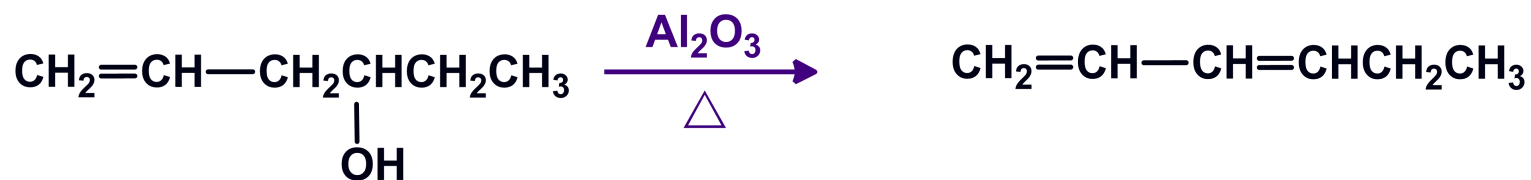
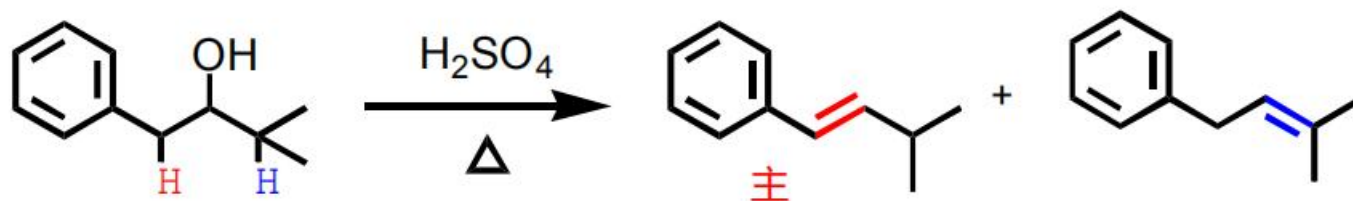
醇的脱水



- 醇脱水得烯，遵循札依采夫规则。



- 优先生成共轭体系。



醇的氧化

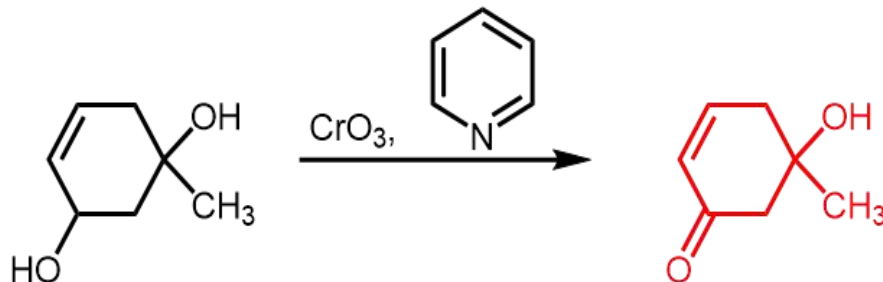


$\text{KMnO}_4 / \text{H}_2\text{SO}_4 / \text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$

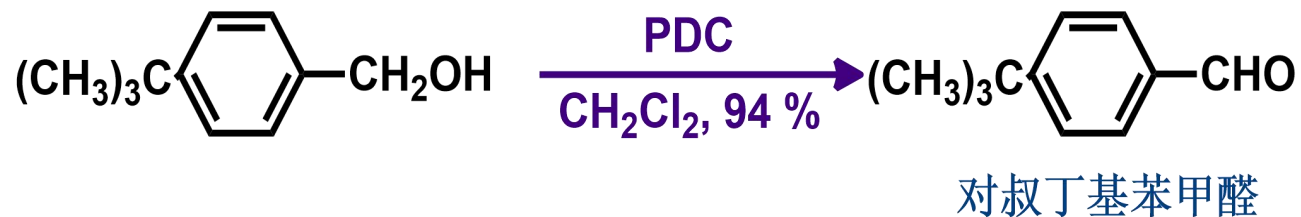
伯醇被氧化成酸；
CCO $\xrightarrow[\text{H}_2\text{SO}_4]{\text{KMnO}_4}$ CC(=O)O

仲醇被氧化为酮；
CC(C)O $\xrightarrow[\text{H}_2\text{SO}_4]{\text{KMnO}_4}$ CC(=O)C

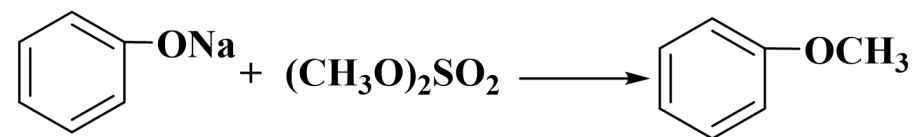
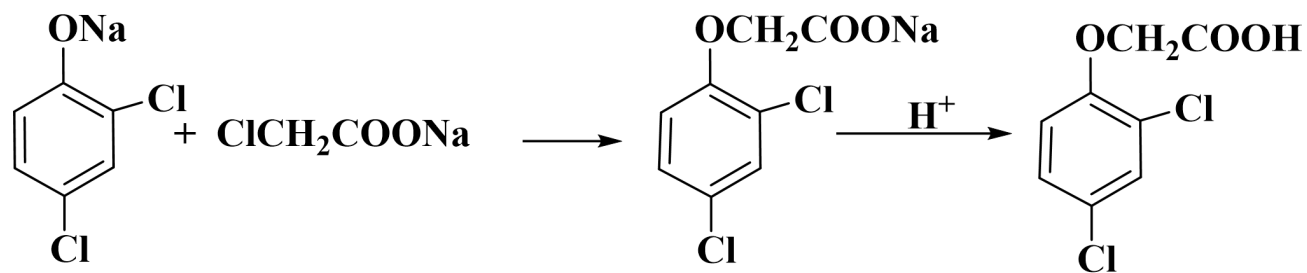
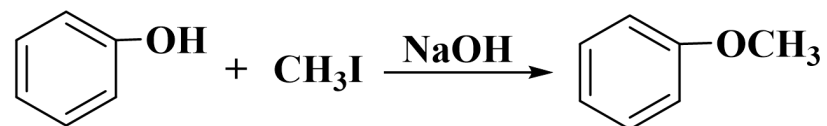
叔醇难以被氧化。
CC(C)(C)O $\xrightarrow[\text{H}_2\text{SO}_4]{\text{KMnO}_4}$?



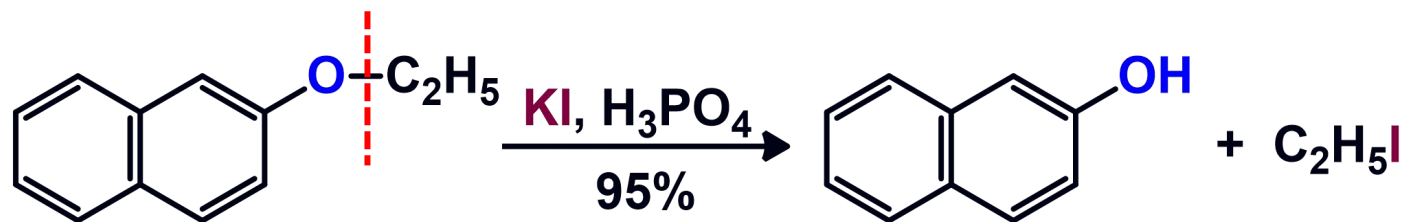
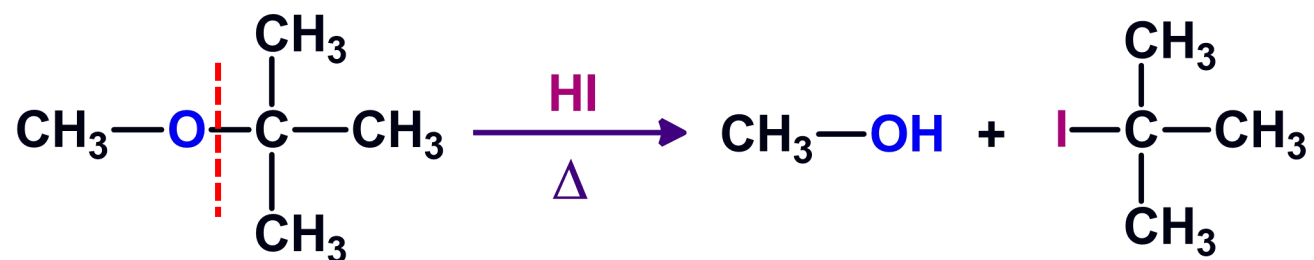
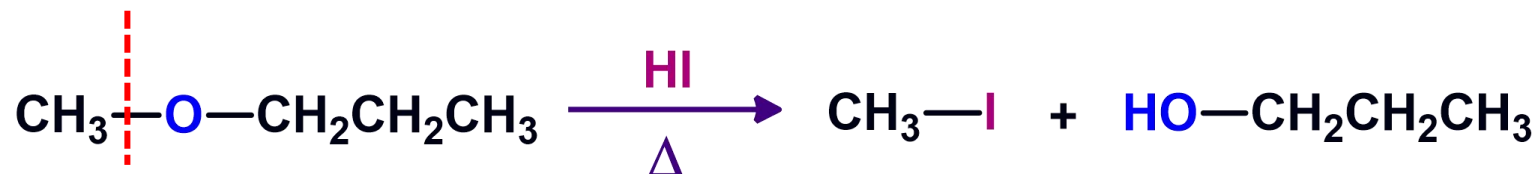
醇--选择性氧化



醚：合成反应



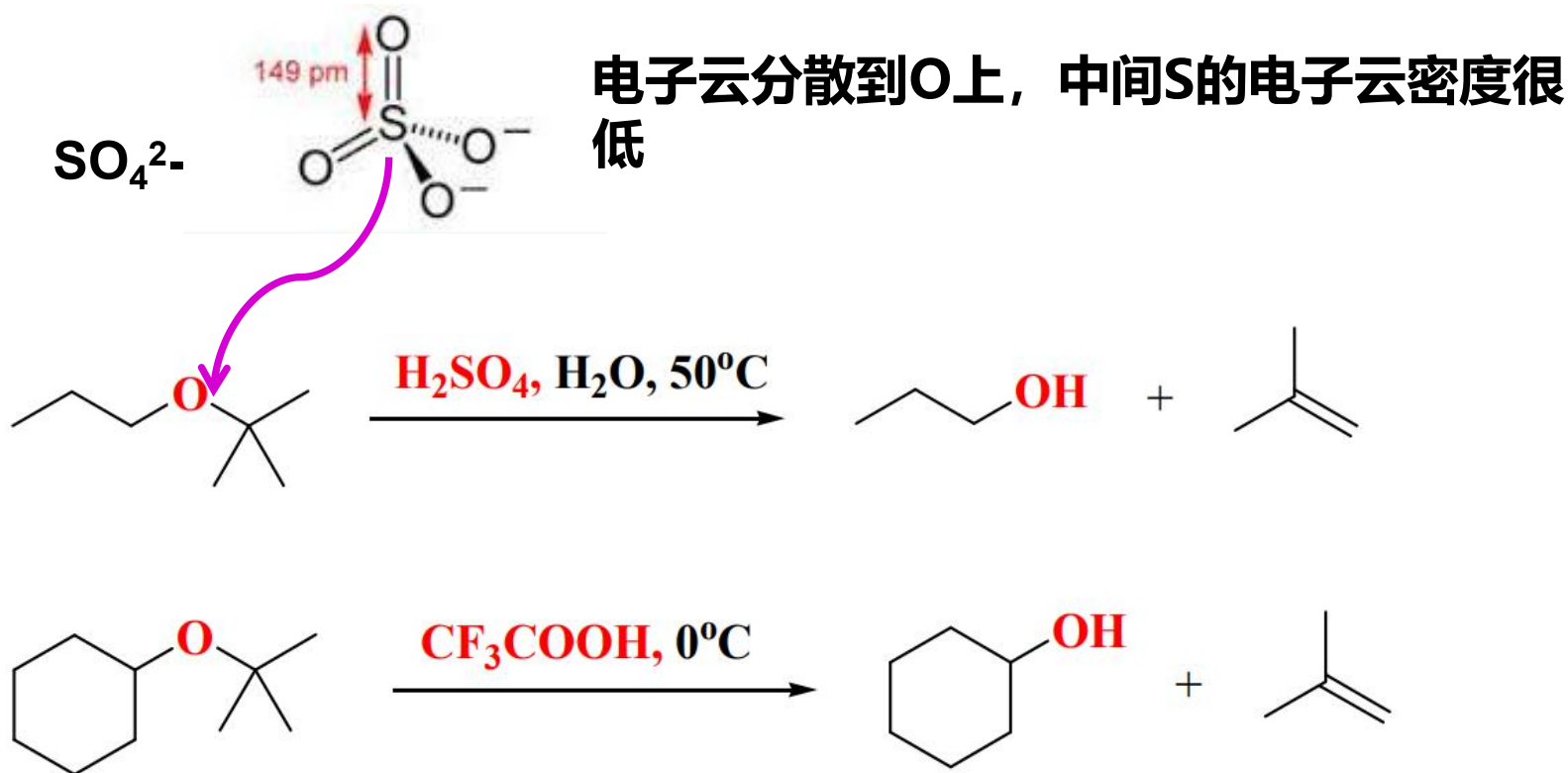
醚：醚键开裂（取向）



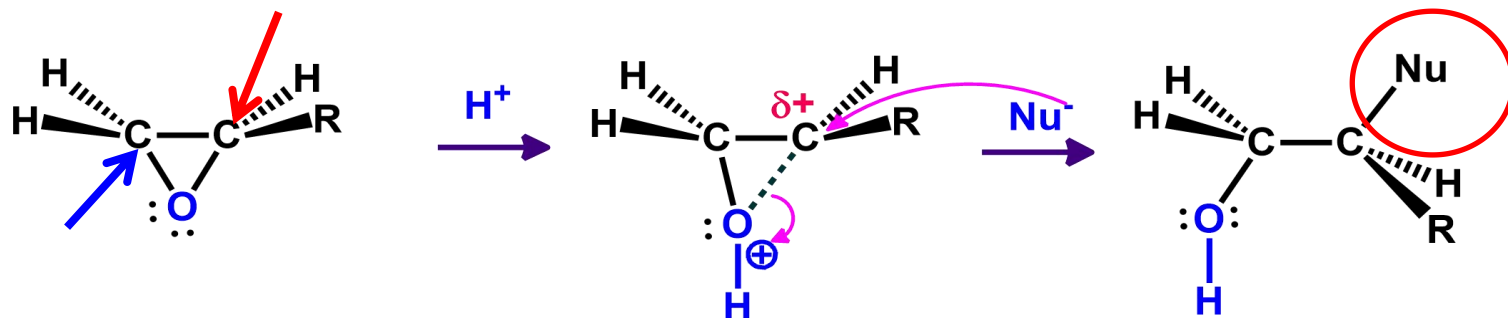
醚：醚键开裂（消除）



如果**Nu-亲核能力很弱**（无法形成C-Nu键），
生成的叔碳正离子主要**发生消除**反应，生成**烯烃**：



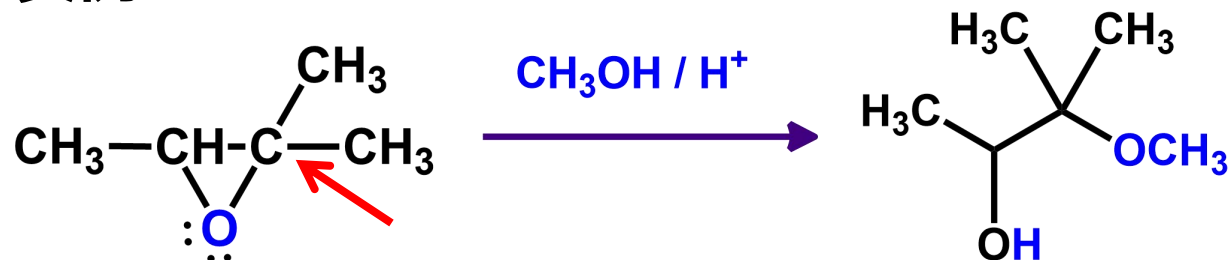
环氧：酸性开环 (C^+ 的稳定性)



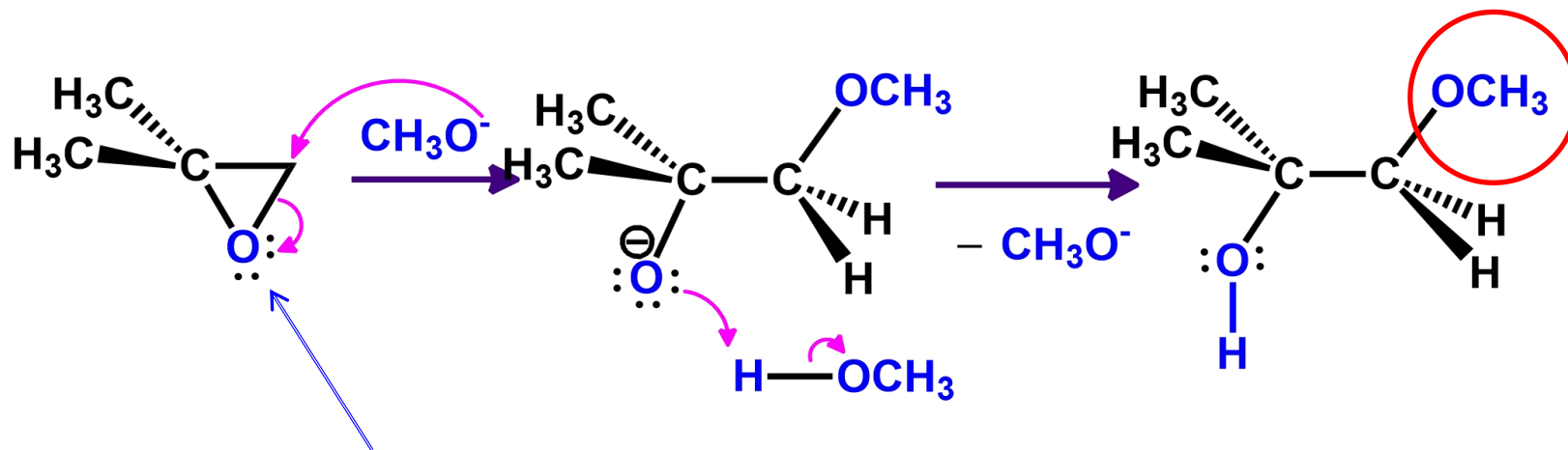
反应取向： Nu^- 优先进攻取代较多的C，类 S_N1

立体化学：反式开环

实例：



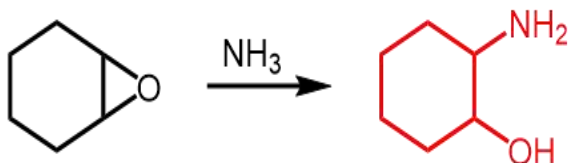
环氧：碱性开环（空间因素）



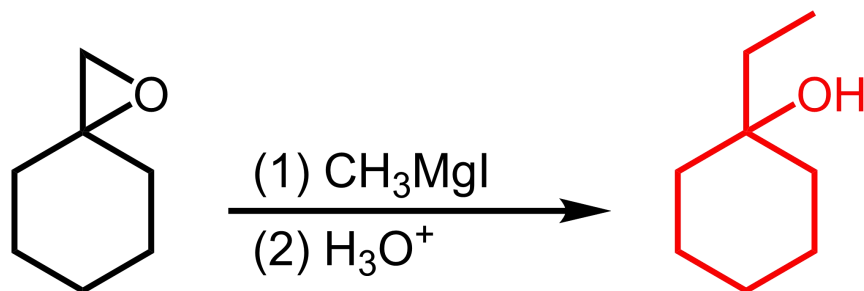
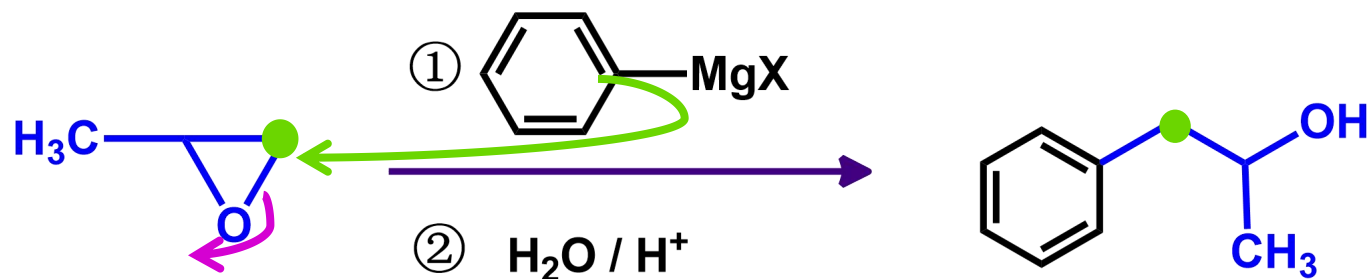
位阻小有利

反应取向：Nu⁻优先进攻取代较少的C，有S_N2性质

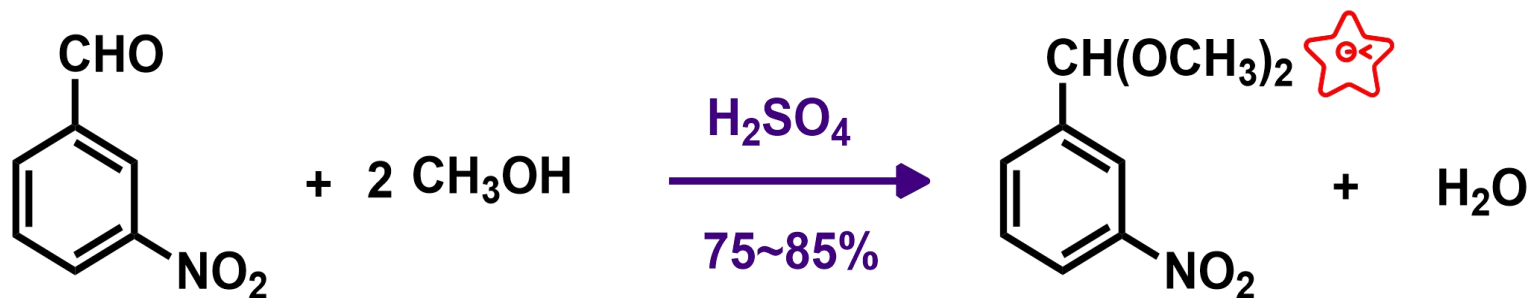
立体化学：反式开环



环氧：与 Grignard 试剂（碱性）

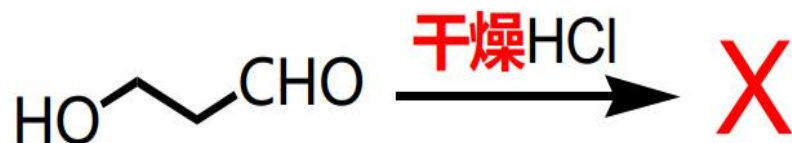


醛酮：与醇缩合（保护）



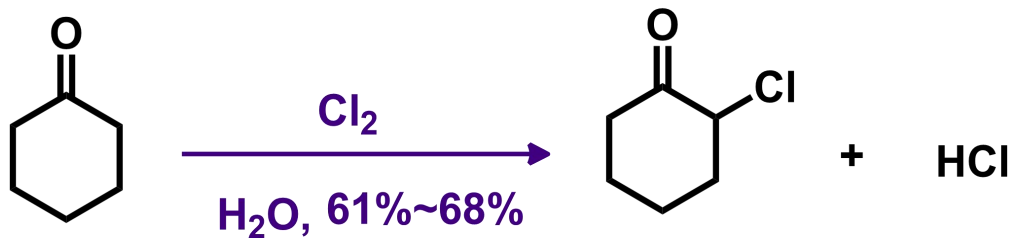
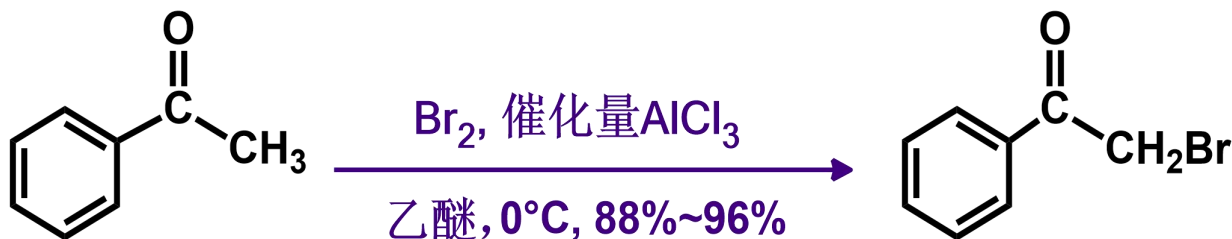
一般用乙二醇代替一元醇，生成**五元环状缩醛或缩酮**

半缩醛：优先生成5/6员环



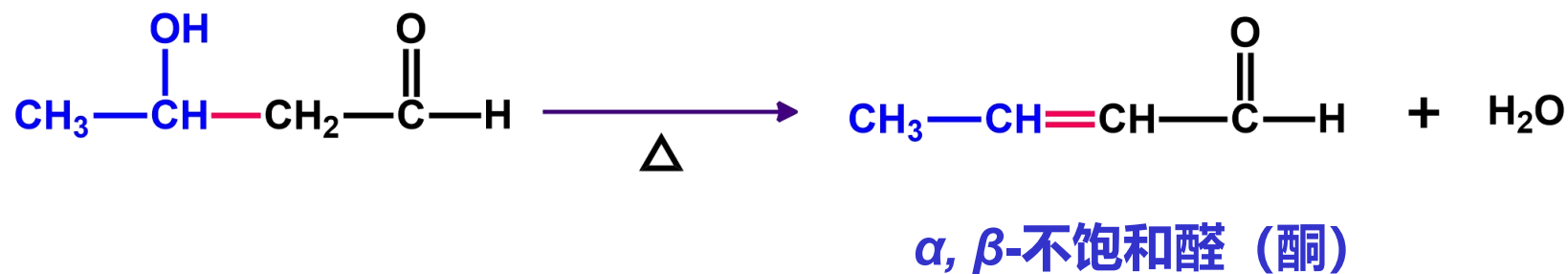
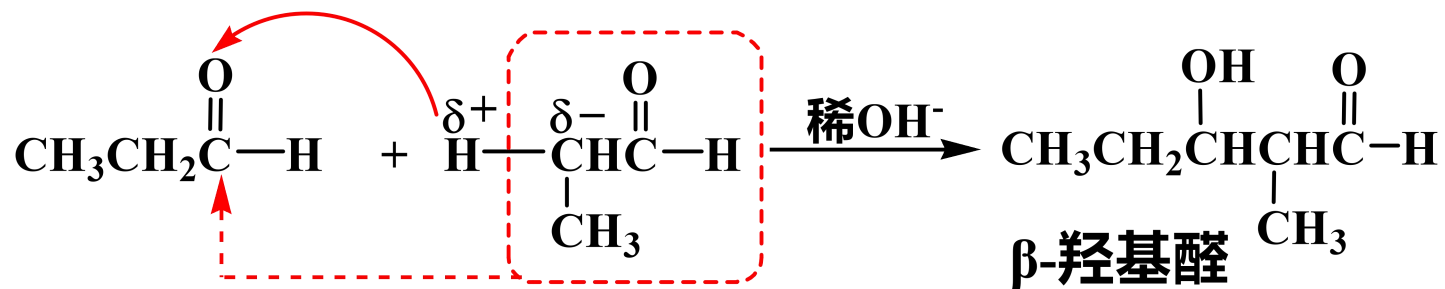
醛：卤化反应

醛、酮分子中的 α -氢原子，在碱或酸的催化下，容易被卤素取代，生成 α -卤代醛、酮。例如：

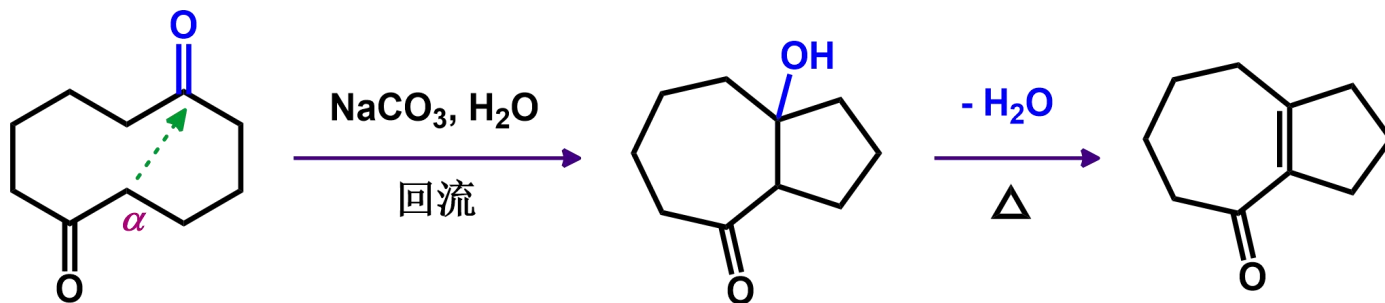
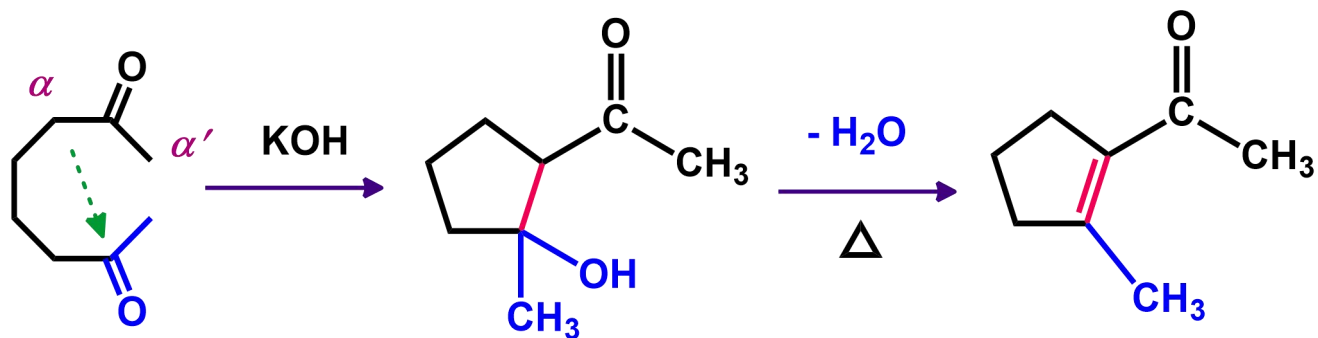


Response	Percentage
Doing a good job	45%
Not doing a good job	55%

羟醛缩合

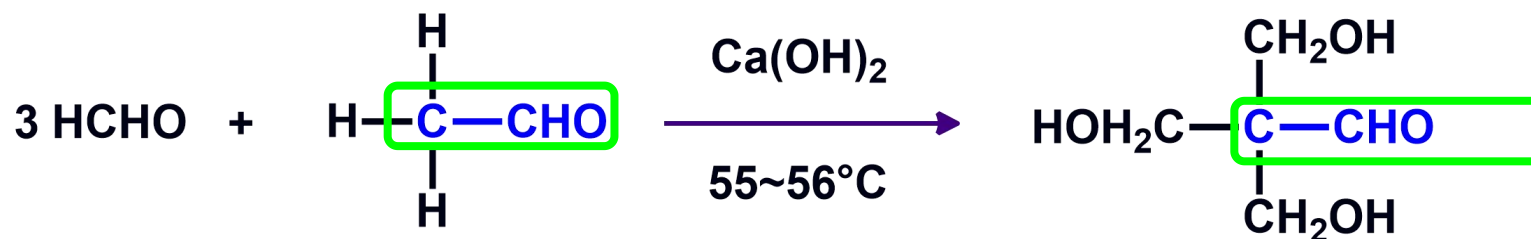


羟醛缩合——分子内

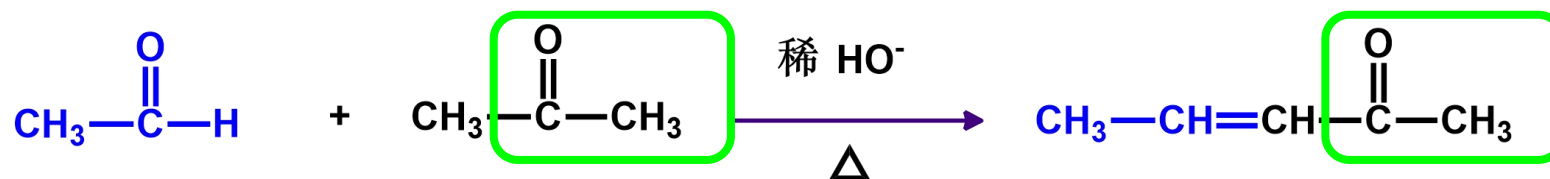


羟醛缩合：交叉缩合

一种没有 α -H，有 α -H的醛酮发生缩合

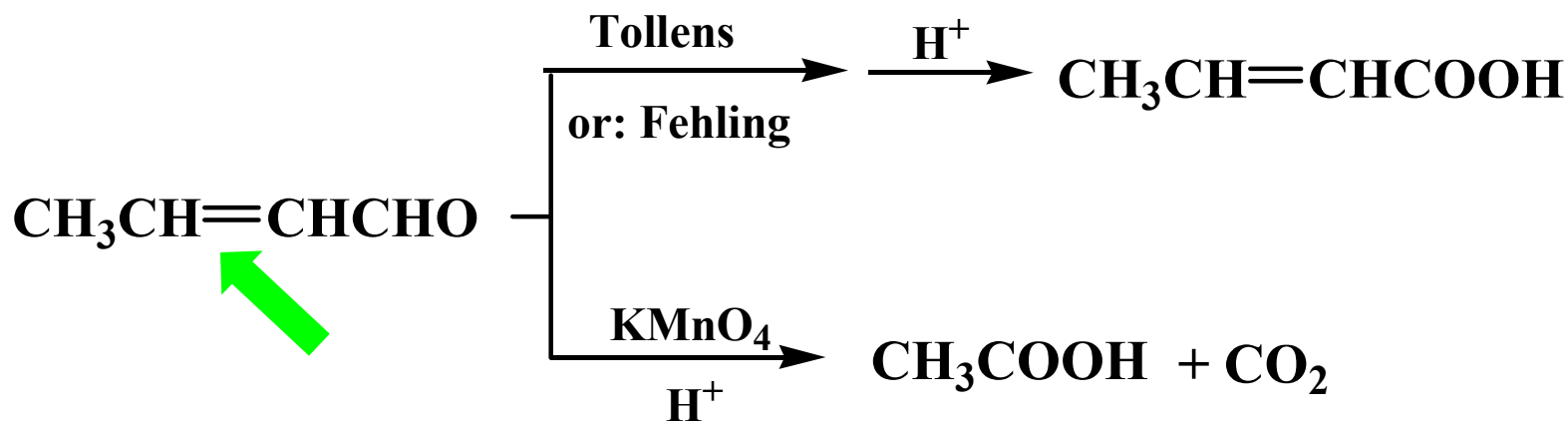


两种醛、酮都有 α -H（一般醛出羰基、酮出 α -H）



醛的氧化:

【1】 Tollens、Fehling试剂均不能氧化C=C键，为选择性氧化剂。

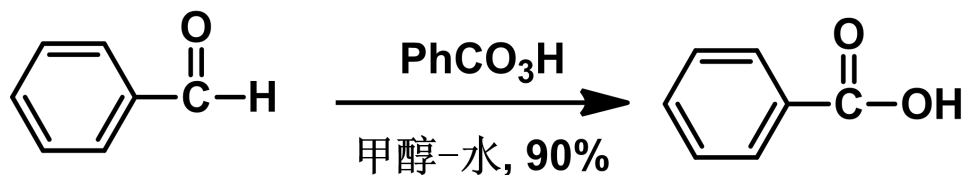


【2】 Fehling试剂用于鉴别脂肪醛与芳香醛（不能氧化芳香醛）

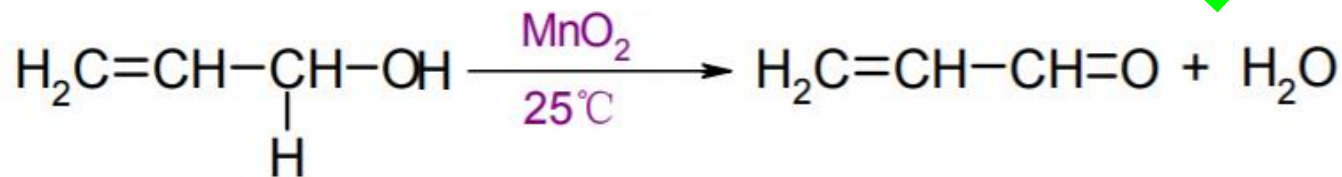
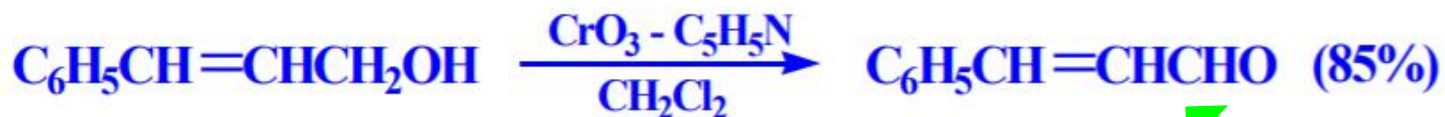
【3】 Tollens、Fehling试剂，鉴定醛和酮（都不与酮反应）

强氧化剂:

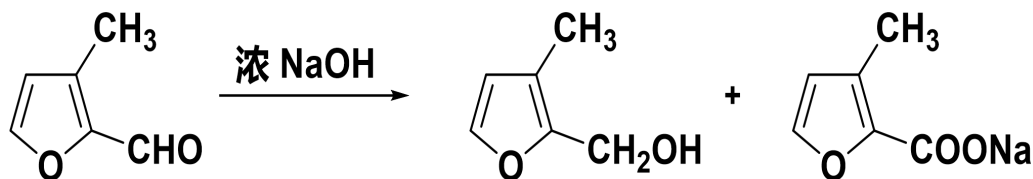
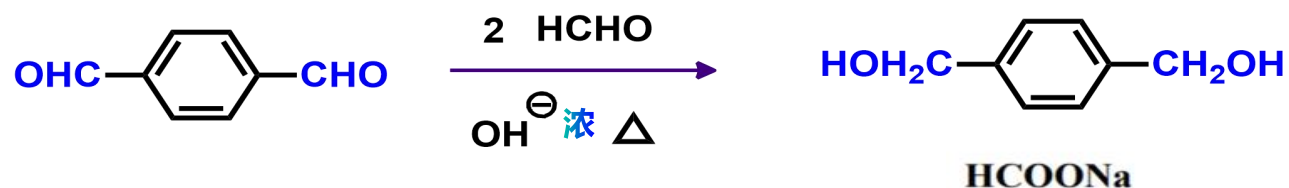
H_2O_2 、 RCO_3H 、 KMnO_4 、 CrO_3 氧化醛被为羧酸



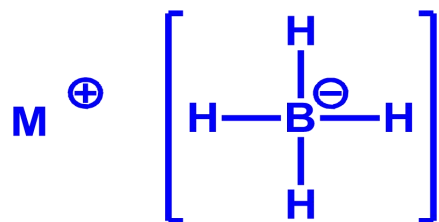
弱氧化剂: 醇氧化为醛, 不再被氧化



坎尼扎罗：甲醛被氧化（注意条件）

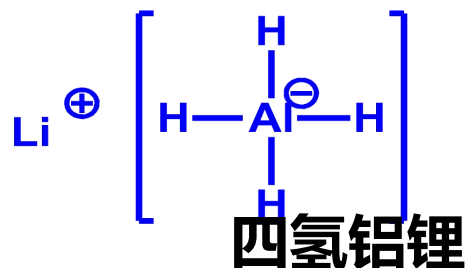


金属氢化物(还原)



硼氢化钠(钾)

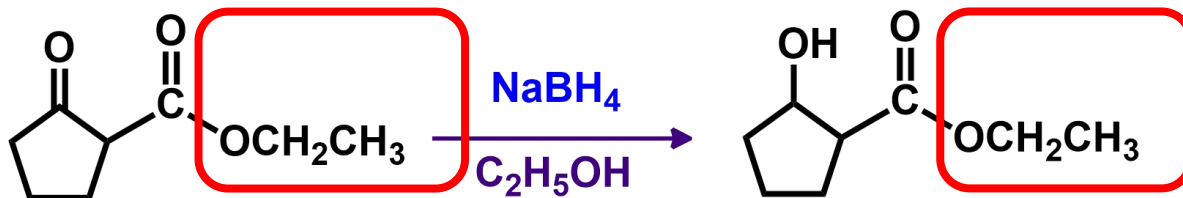
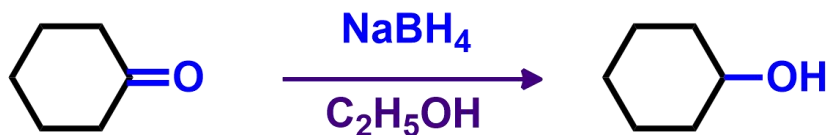
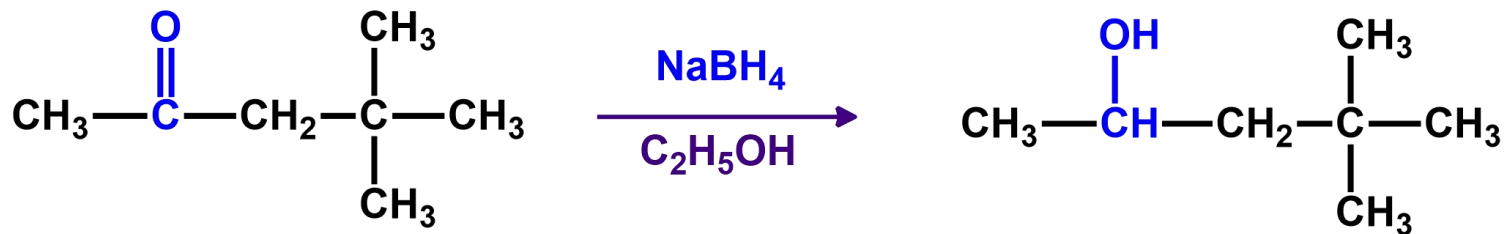
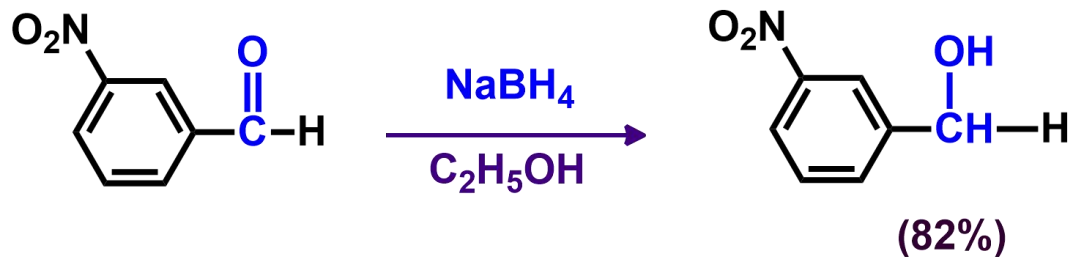
$M = Na, K$



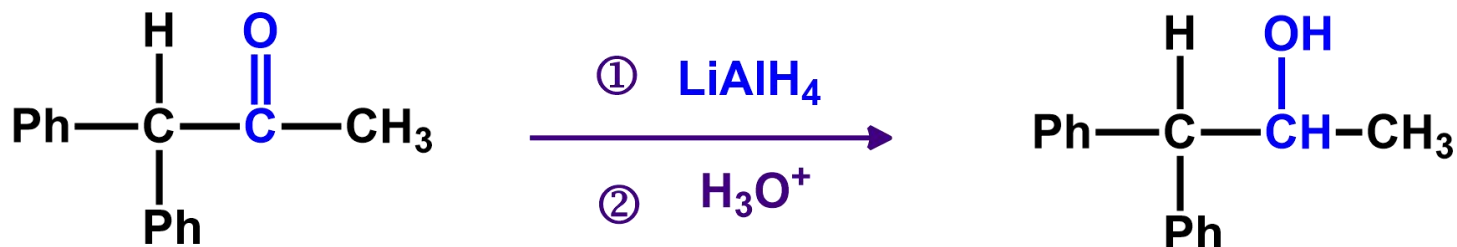
还原特点:

- $NaBH_4$ 可还原: 醛、酮
- $LiAlH_4$ 可还原: 醛、酮、羧酸、酯、酰胺、腈、硝基
- 都不能还原: 碳碳双键、叁键

NaBH₄ : 不能还原酯



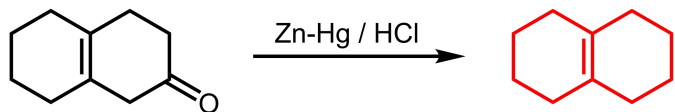
LiAlH_4 : H^- 还原能力极强, 但不能与 $\text{C}=\text{C}$ 反应



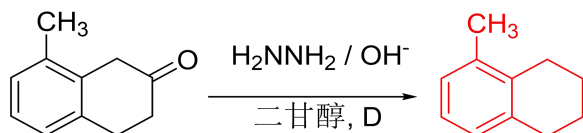
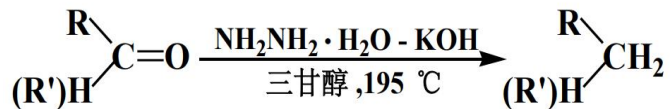
羰基：还原为亚甲基



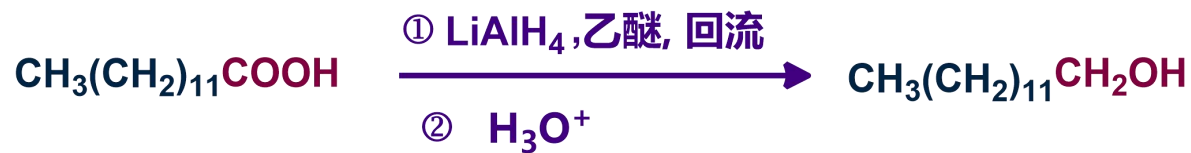
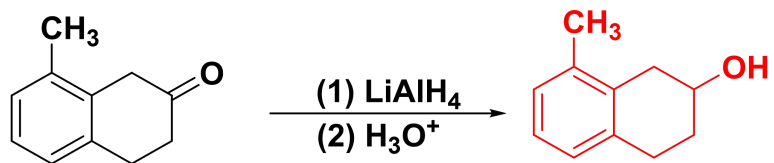
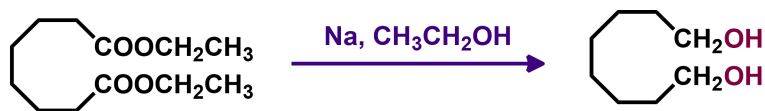
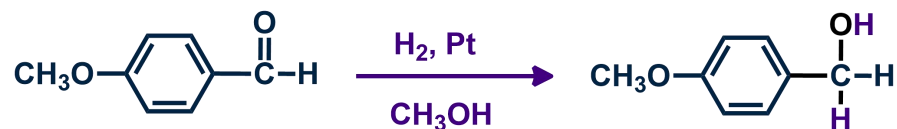
克莱门森 (Clemmensen) 还原—酸性条件



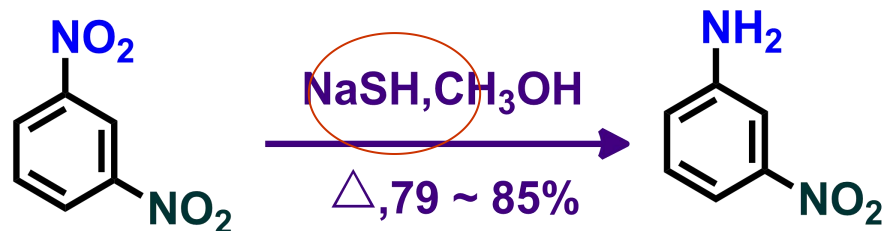
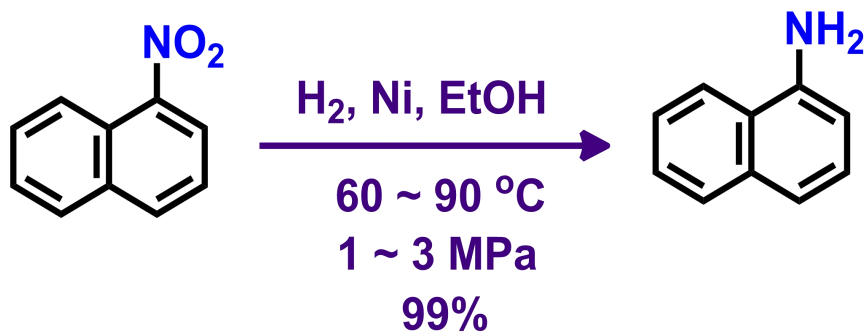
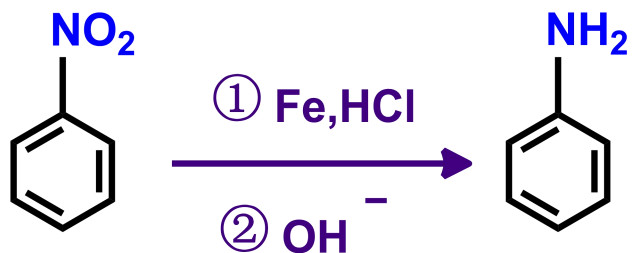
沃尔夫-吉斯尼尔-黄鸣龙还原—碱性条件



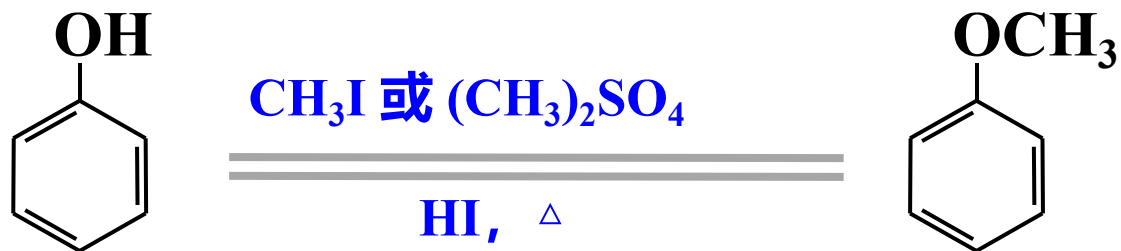
醛、酮、羧酸和羧酸衍生物还原



硝基：还原成为NH₂-- 合成题

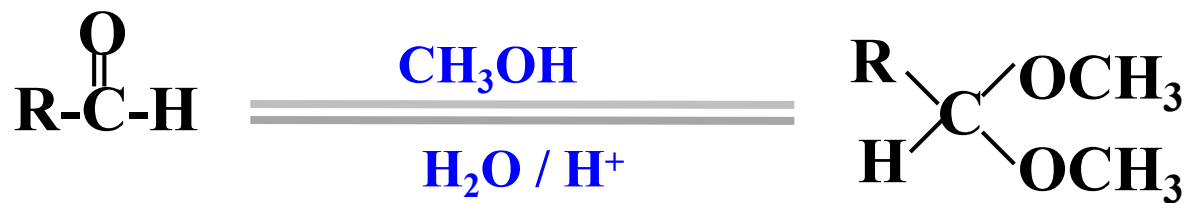
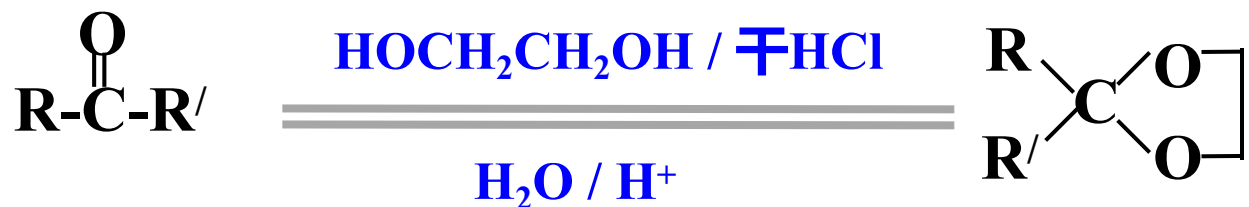


保护醇/酚类



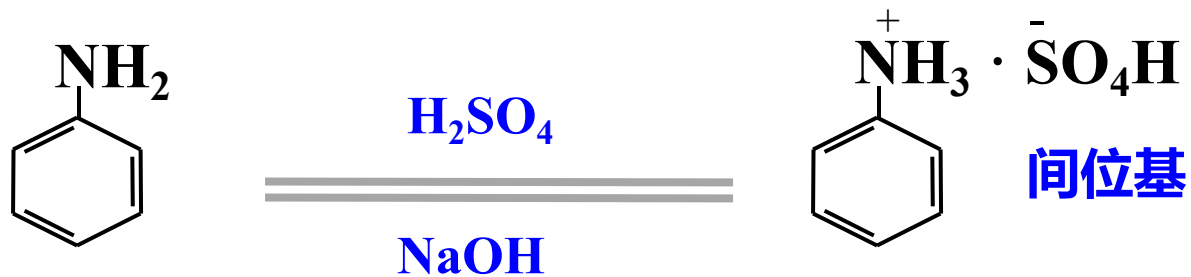
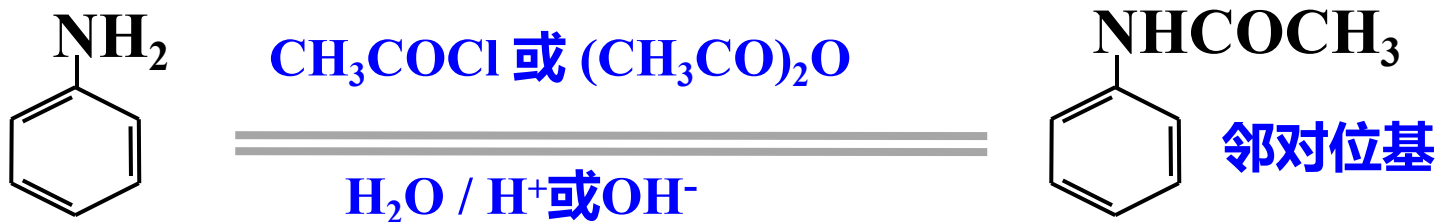
苯甲醚：对碱、氧化剂、还原剂等稳定，但对酸不稳定。

醛、酮、羰基的保护

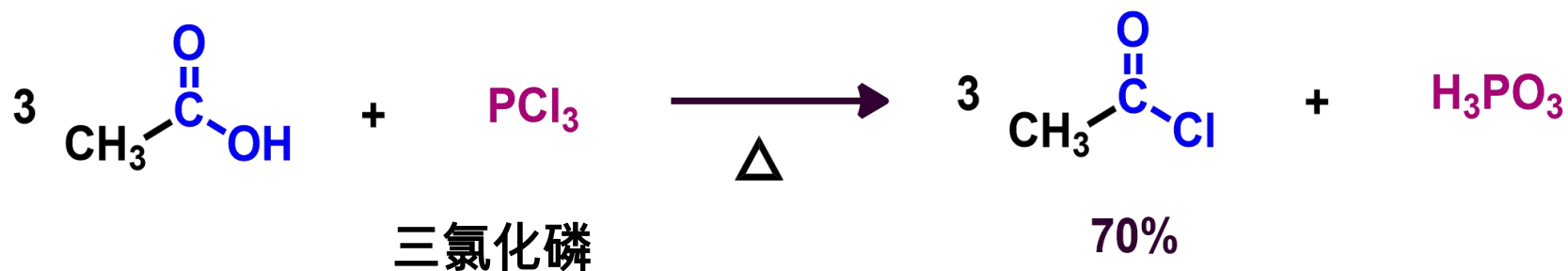
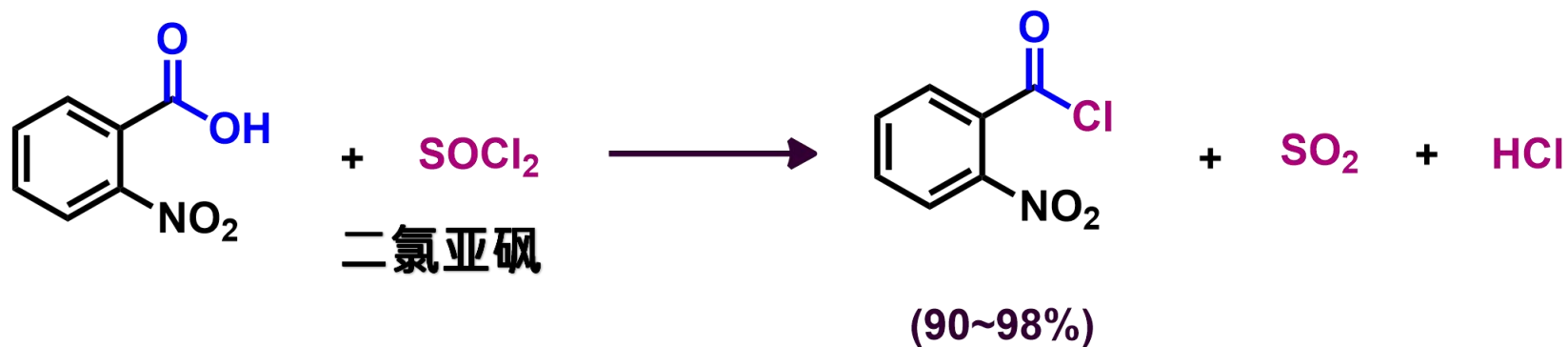


缩醛（酮）：对还原剂、氧化剂、碱、格氏试剂稳定，但对酸不稳定。

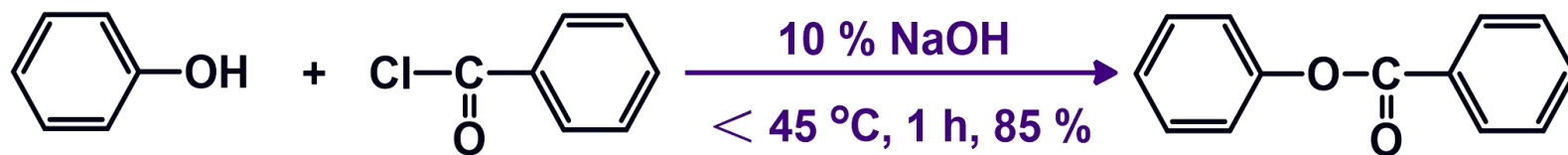
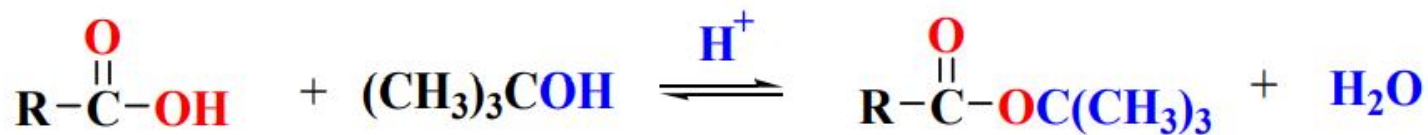
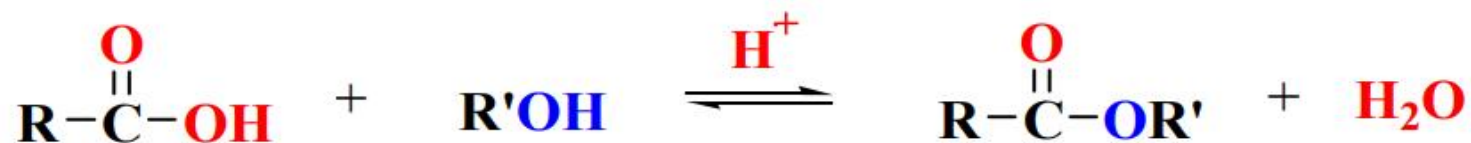
氨基的保护



合成酰氯

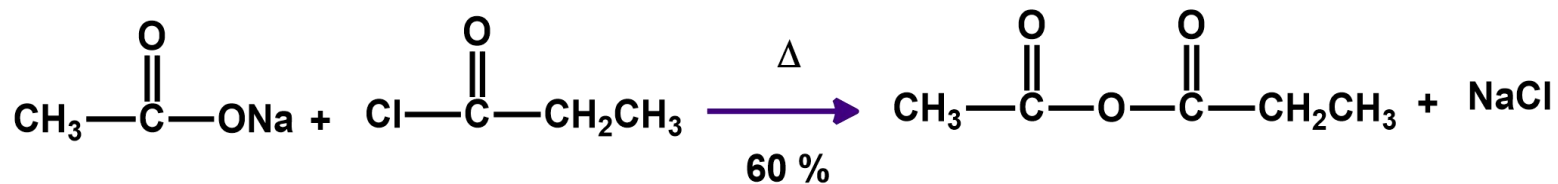
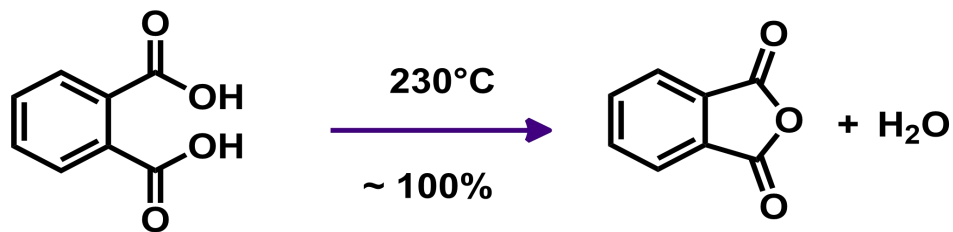
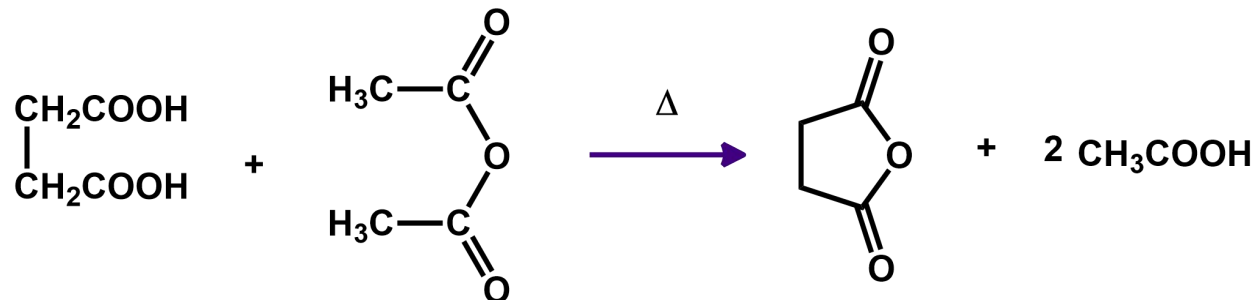
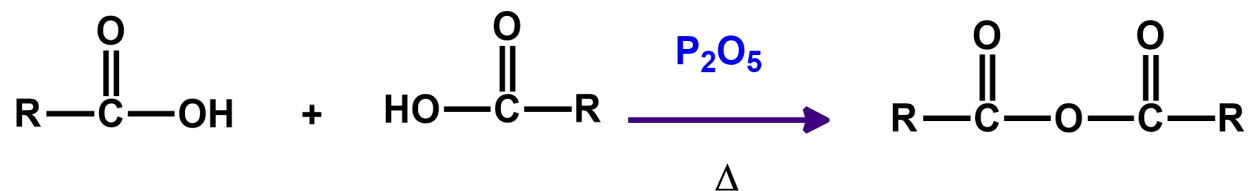


合成酯

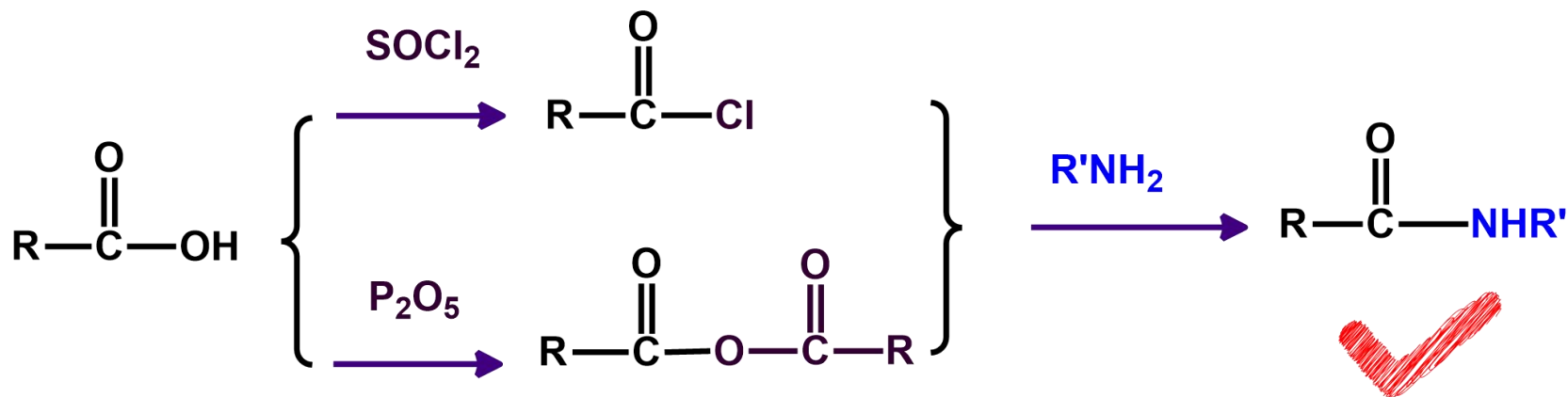
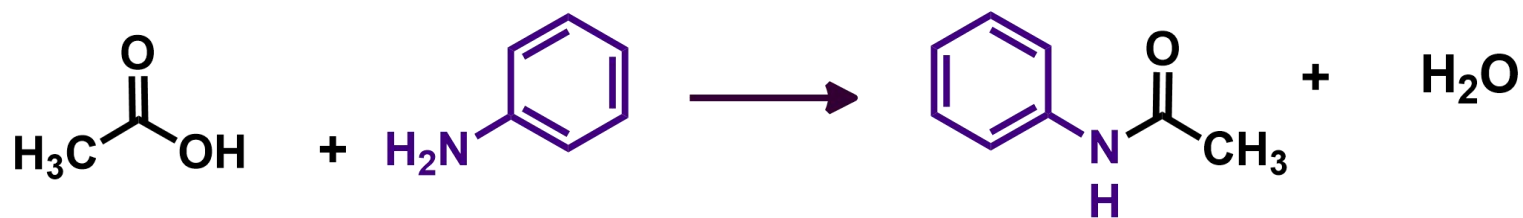


苯甲酸苯酯

合成酸酐

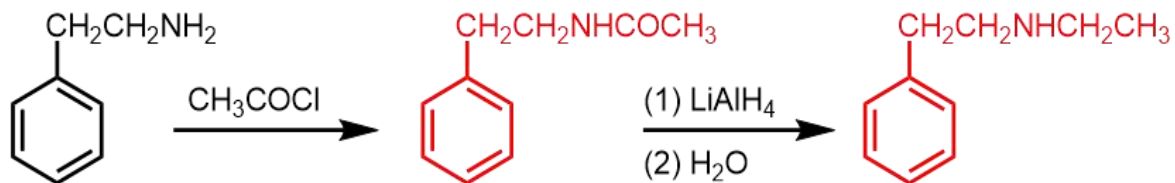
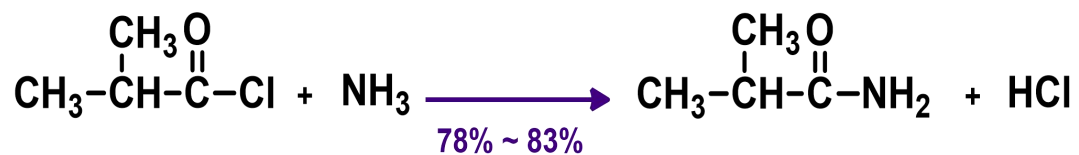
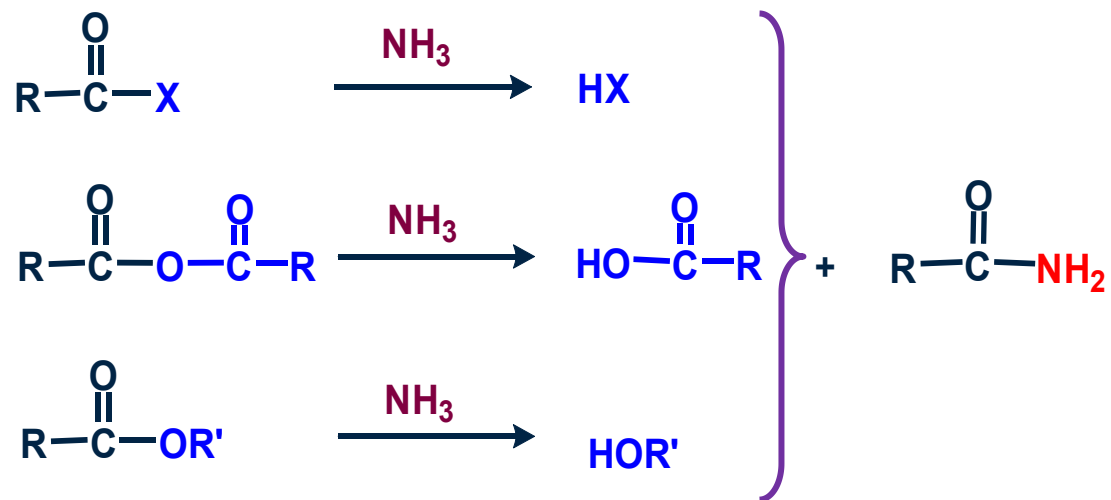


合成酰胺

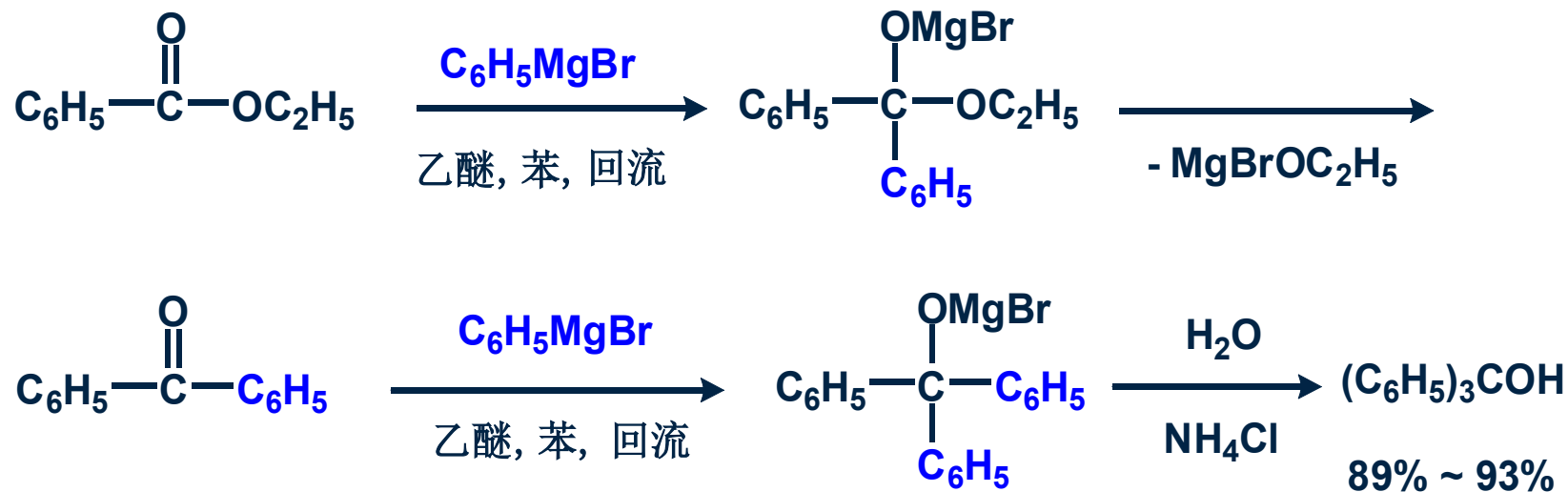


羧酸衍生物

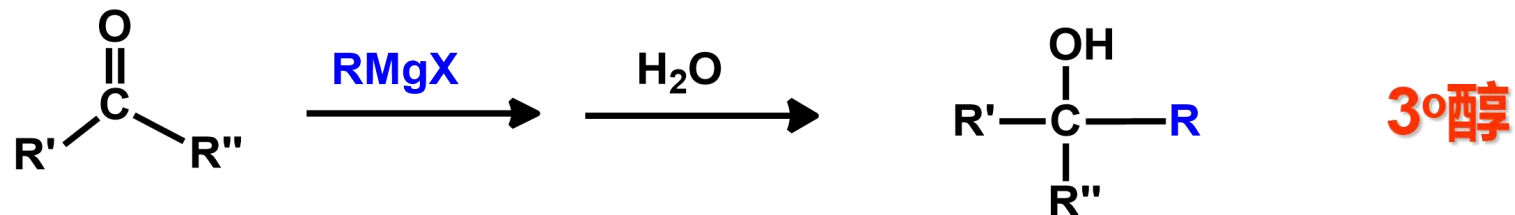
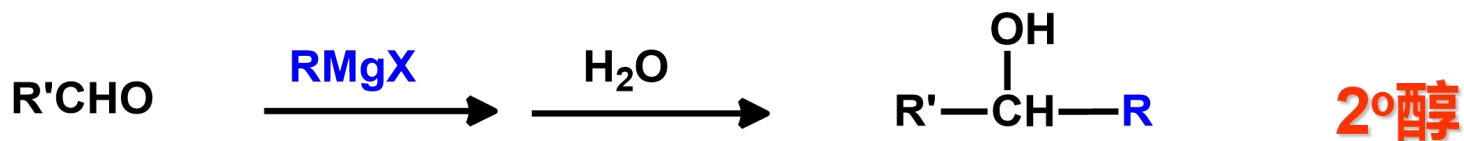
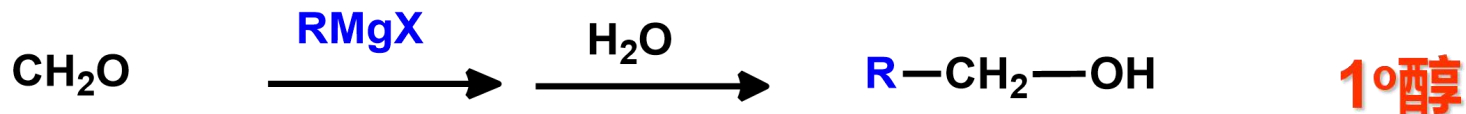
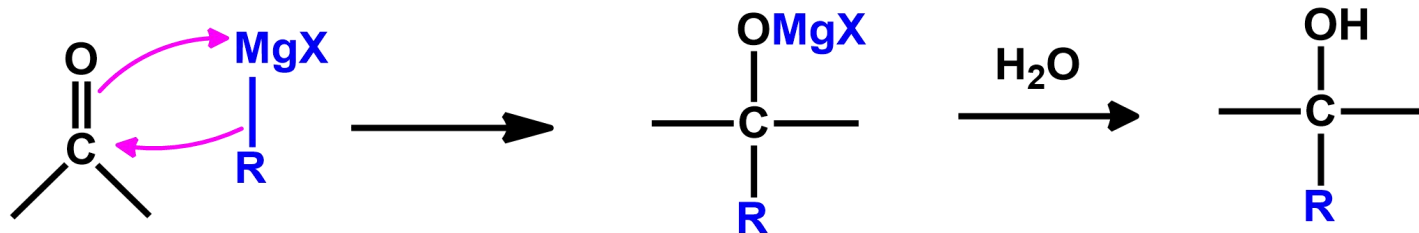
合成酰胺(氨解)



合成叔醇

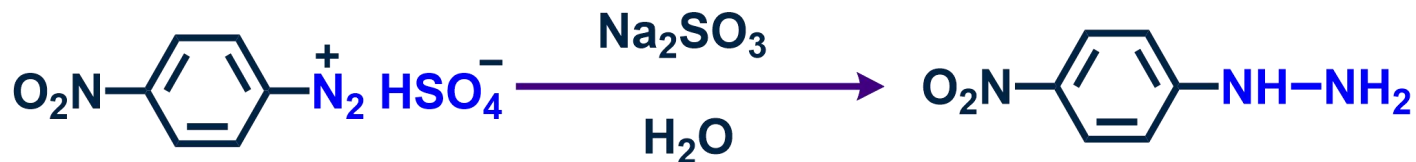
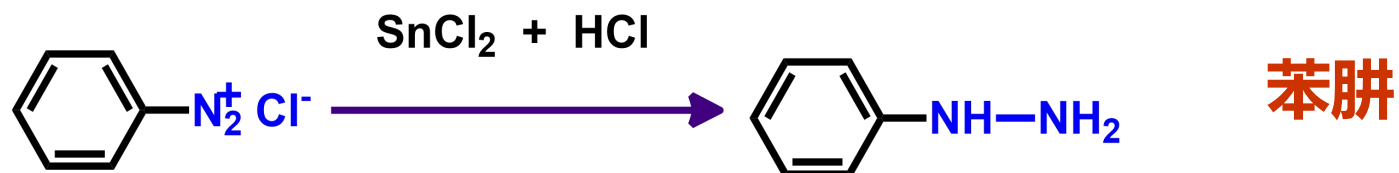


与Grignard试剂的加成

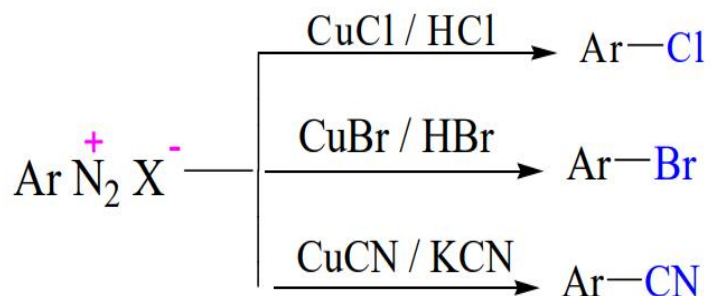


-N=N 被还原

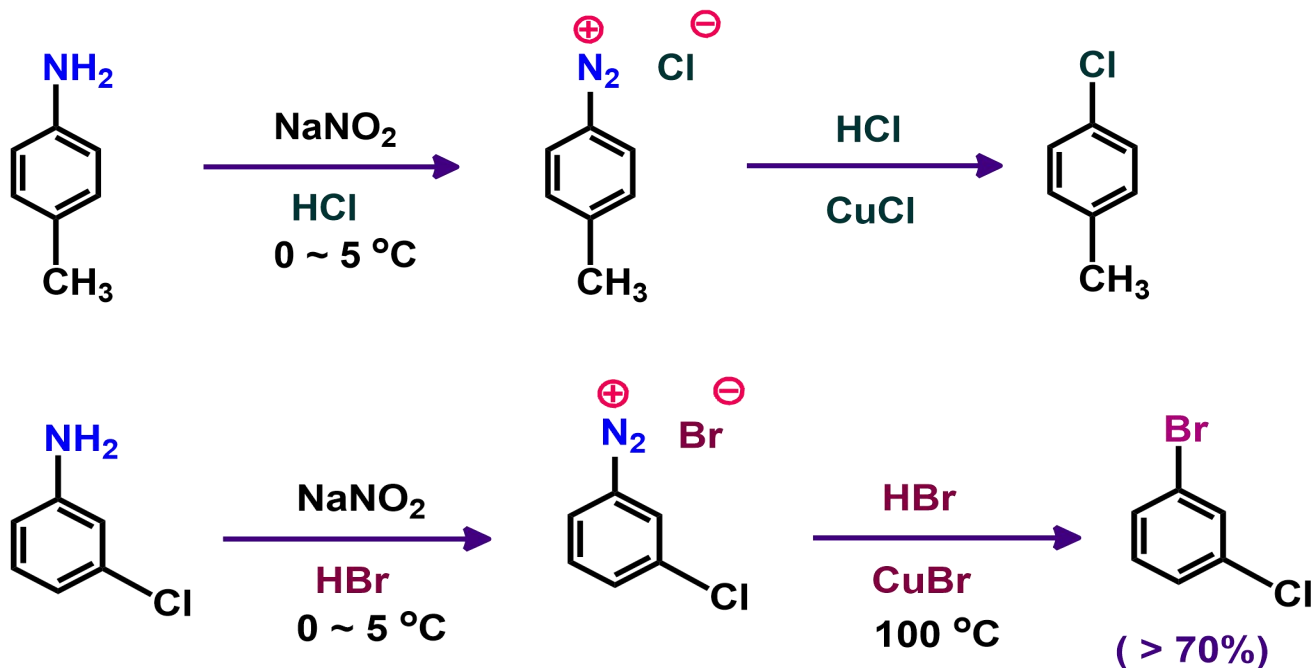
$\text{SnCl}_2 + \text{HCl}$, NaHSO_3 , Na_2SO_3 , SO_2 等还原为芳基肼。



重氮基受热被置换成Cl, Br、CN



在亚铜盐的催化作用下



合成酰胺(氨解)

